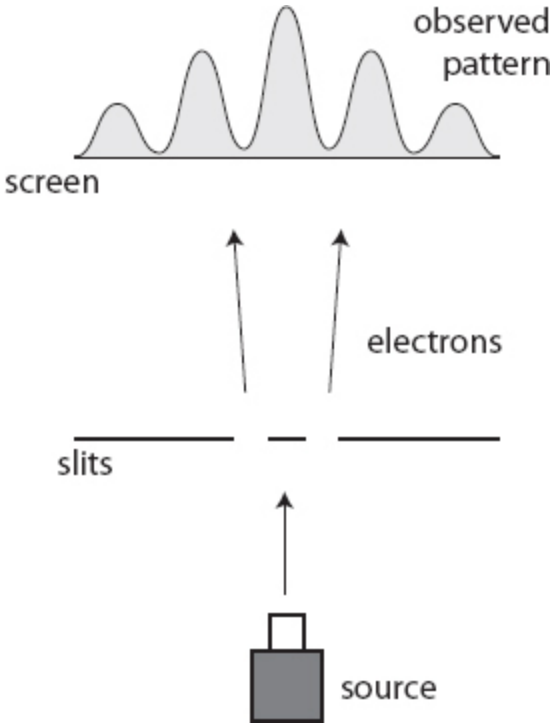


Let’s introduce one additional wrinkle. Imagine that we set up little detectors at each slit, so we can tell whether an electron goes through it. That will settle this crazy idea that an electron can travel through two slits once and for all.

You should be able to figure out what we see. The detectors don’t measure half of an electron going through each of the two slits; they measure a full electron going through one, and nothing through the other, every time. That’s because the detector acts as a measuring device, and when we measure electrons we see particles.

But that’s not the only consequence of looking at the electron as it passes through the slits. At the screen, on the other side of the slits, the interference pattern disappears, and we are back to seeing two bands of marks made by the detected electrons, one for each slit. With the detectors doing their job, the wave function collapses as the electron goes through the slits, so we don’t see interference from a wave passing through both slits at once. When we’re looking at them, electrons behave like particles.